

VIRTUAL PLCs Are Coming

As industries embrace digital transformation, virtual PLCs will be key to boosting efficiency, cutting costs, and enabling innovation. The future of industrial automation is clearly software-driven.



The author, **VINAYAK RAMACHANDRA ADKOLI**, is BE in Industrial Production and has served as a lecturer in three different polytechnics for ten years. He is also a freelance writer and cartoonist.



Programmable logic controllers (PLCs) have long served as the backbone of industrial automation, reliably managing machinery and processes across sectors. Traditionally hardware-based, PLCs were built for specific control tasks. Now, with advances in computing and cloud technologies, a new paradigm has emerged—Virtual PLCs (vPLCs). These software-based controllers bring greater flexibility, scalability, and connectivity, reshaping the future of industrial automation.

What are virtual PLCs?

A virtual PLC (vPLC) is a software-based controller that performs the same

functions as a traditional hardware PLC but operates on standard computing platforms. Instead of being tied to dedicated hardware, vPLCs run on industrial PCs, edge devices, or cloud servers. This virtualisation enables industries to deploy, manage, and scale control systems more efficiently while reducing reliance on proprietary hardware.

How do vPLCs work?

Virtual PLCs (vPLCs) utilise a combination of industrial computing, virtualisation, and cloud technologies to function effectively. Unlike traditional PLCs that rely on dedicated hardware, vPLCs operate on general-purpose